

Q) Find the Fourier Inverse of $\frac{3i\omega}{2+i\omega}$

Ans) $3\delta(t) - 6e^{-2t}u(t)$ (Hint : $3 - \frac{6}{2+i\omega}$)

Q) Find the real and imaginary part of Fourier Transformation of $e^{a|t|}$

Ans) Real part = $\frac{2a}{a^2+\omega^2}$ and Imaginary part = 0

Q) Find the Fourier transformation of $e^{-3|t|}\sin(2t)$

Hint : Use Modulation Theorem

Q) $\mathcal{F} \left(\frac{1}{(3t)^2+1} \right)$

Hint : Use symmetric property

Ans) $\frac{\pi}{3} e^{-\left|\frac{\omega}{3}\right|}$

Q) Solve $y' + 2y = e^{-t}u(t)$

Ans) $e^{-t}u(t) + e^{-2t}u(t)$

Q) Prove that $\mathcal{F} [t^{n-1}e^{-at}u(t)] = \frac{(n-1)!}{(a+iw)^n}$